CHAPTER 1 — INTRODUCTION

1.1 Company Profile

DocuMind is an AI-based web application . It is designed to simplify document understanding by allowing users to upload documents and interact with them using natural language queries.

The system integrates modern web technologies along with artificial intelligence to provide intelligent responses based on the content of uploaded documents. It uses a user-friendly interface where users can register, log in, upload files, and ask questions.

DocuMind demonstrates the practical implementation of technologies such as React, Node.js, Express, MongoDB, and OpenAI API. The project aims to showcase how AI can be used to improve document analysis and information retrieval in real-world applications.

1.2 Problem Statement

In today's digital environment, users often need to extract information from lengthy documents. Manually reading and searching for specific information is time-consuming and inefficient. Existing systems lack an interactive way to query documents directly. Therefore, there is a need for a system that allows users to upload documents and interact with them using natural language queries to obtain relevant information quickly.

1.3 Objectives of Proposed System

The main objectives of the system are:

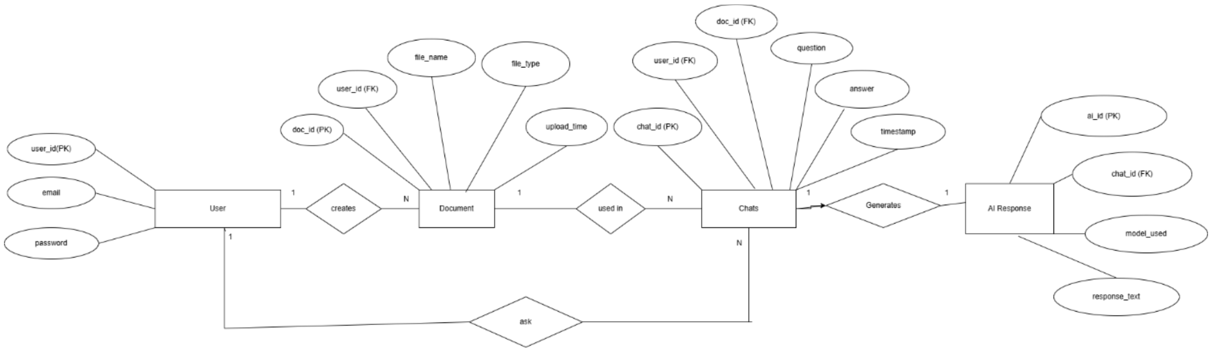
- To develop a web-based application for document interaction
- To allow users to upload documents easily
- To enable users to ask questions based on uploaded documents
- To generate intelligent answers using AI
- To provide a simple and user-friendly interface

1.4 Scope of Proposed System

The system is designed for users who need quick insights from documents. It supports document upload and question-answer interaction using AI. The application is suitable for academic, research, and general-purpose document analysis. However, the system currently supports limited document formats and does not store long-term chat history.

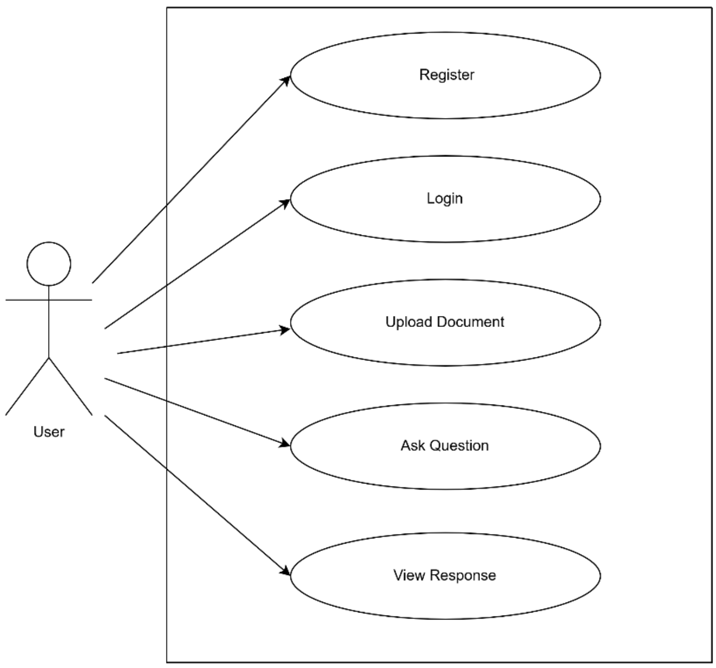
CHAPTER 2 — ANALYSIS AND DESIGN

3.1 Entity Relationship Diagram (ERD):



The ER diagram represents the structure of the system and the relationships between entities such as User, Document, Chat, and AI Response. A user can upload multiple documents and ask multiple questions. Each question generates an AI response. The relationships are maintained using unique identifiers.

3.2 Business Use Case Diagram



The use case diagram illustrates the interaction between the user and the system. The user can register, log in, upload documents, ask questions, and view AI-generated responses. It represents the functional flow of the application.

3.3 Table Structure / Data Dictionary

USER TABLE

Table Name: User

Field Name	Type	Description
user_id	int	Primary Key
email	varchar	User email
password	varchar	Encrypted password

DOCUMENT TABLE

Table Name: Document

Field Name	Type	Description
doc_id	int	Primary Key
user_id	int	Foreign Key
file_name	varchar	Name of file
file_type	varchar	File type
upload_time	datetime	Upload timestamp

CHAT TABLE

Writing

Table Name: Chat

Field Name	Type	Description
chat_id	int	Primary Key
user_id	int	Foreign Key
doc_id	int	Foreign Key
question	text	User question
answer	text	AI answer
timestamp	datetime	Time of interaction

AI RESPONSE TABLE

Table Name: AIResponse

Field Name	Type	Description
ai_id	int	Primary Key
chat_id	int	Foreign Key
model_used	varchar	AI model name
response_text	text	Generated response

CHAPTER 3 — TESTING

4.1 Test Cases with Results

TEST CASE TABLE

Test Case ID	Description	Input	Expected Output	Actual Output	Status
TC01	User Registration	Email, Password	User registered successfully	User registered successfully	Pass
TC02	Duplicate Registration	Same Email	Error message	Error message displayed	Pass
TC03	User Login (Valid)	Correct credentials	Login successful	Login successful	Pass
TC04	User Login (Invalid)	Wrong password	Error message	Error message displayed	Pass
TC05	Upload Document	PDF file	File uploaded	File uploaded	Pass
TC06	Ask Question	Valid question	AI response	AI response generated	Pass
TC07	Empty Question	No input	Error message	Error shown	Pass
TC08	AI Response Accuracy	Question based on doc	Relevant answer	Relevant answer	Pass

4.2 Defect Report / Test Log

During testing, some issues were identified and resolved. Initially, file upload errors occurred due to incorrect configuration. There were also issues with API integration, including rate limits and invalid responses. These issues were fixed by correcting API usage and improving error handling. After fixing the defects, the system performed as expected.

CHAPTER 4 — LIMITATIONS OF PROPOSED SYSTEM

The system has some limitations:

- It does not store chat history permanently
- It supports limited file formats
- AI responses depend on external API availability
- Performance may vary based on document size
- Requires internet connection for AI processing

CHAPTER 5 — PROPOSED ENHANCEMENTS

The system can be enhanced in the following ways:

- Add support for multiple document formats
- Implement chat history storage
- Improve response accuracy using advanced NLP techniques
- Integrate vector databases for better document search
- Add user dashboard and analytics

CHAPTER 6 — REFERENCES

- OpenAI API Documentation
- Node.js Documentation
- Express.js Documentation
- MongoDB Documentation
- React.js Documentation

CHAPTER 7 — USER MANUAL

The user manual provides instructions for using the DocuMind system. It explains how to interact with the system step-by-step.

STEPS:

Step 1: Register

Writing

The user opens the application and registers using email and password.

Step 2: Login

Writing

The user logs in using registered credentials.

Step 3: Upload Document

Writing

The user selects and uploads a document file.

Step 4: Ask Question

Writing

The user enters a query related to the uploaded document.

Step 5: View Response

Writing

The system processes the query and displays an AI-generated response.